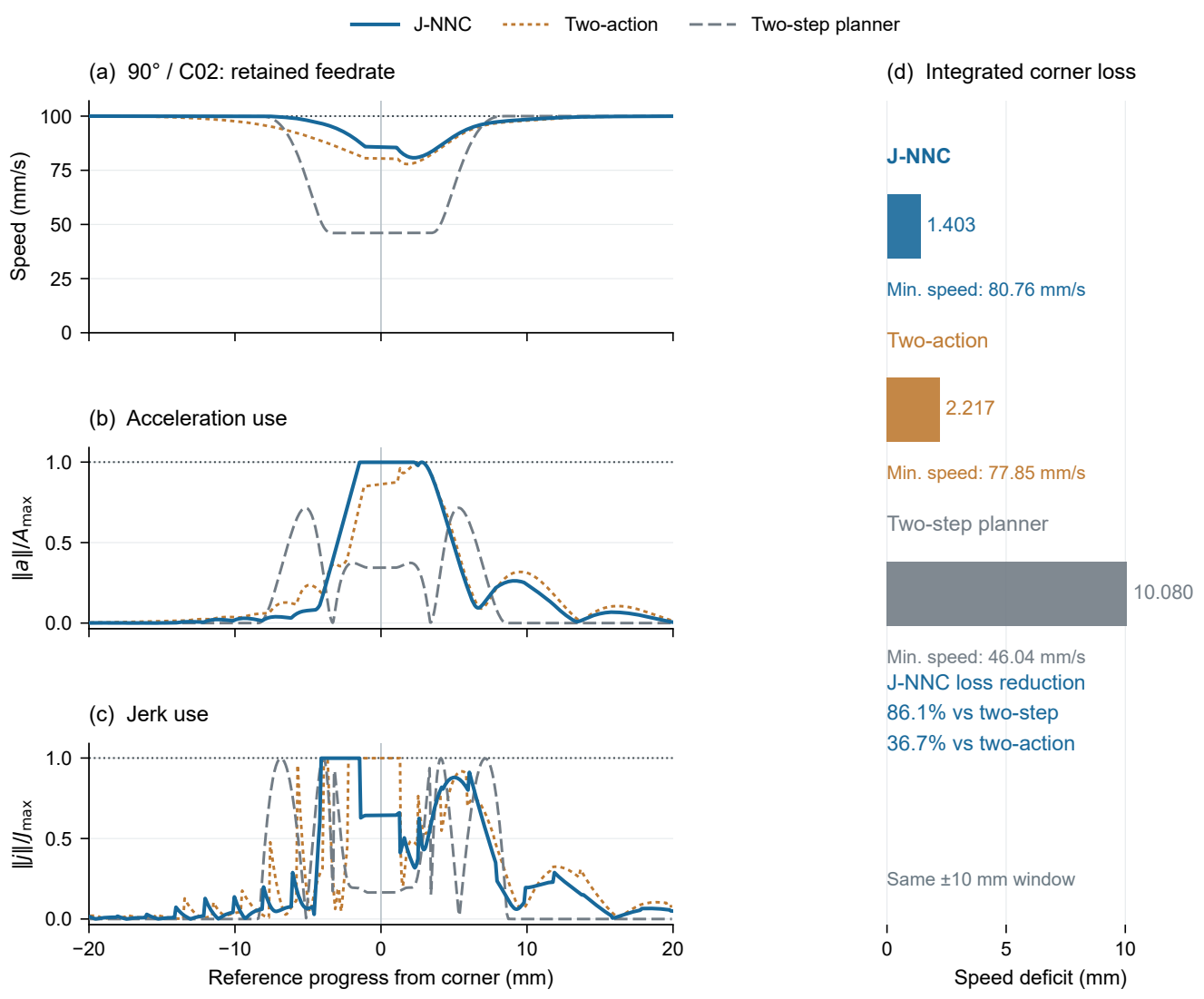

J-NNC
Two-action
Two-step planner
(a) 90° / C02: retained feedrate
Speed (mm/s)
100
75
50
25
0
(b) Acceleration use
$\|a\|/A_{max}$
1.0
0.5
0.0
(c) Jerk use
$\|j\|/J_{max}$
1.0
0.5
0.0
−20
−10
0
10
20
Reference progress from corner (mm)
(d) Integrated corner loss
J-NNC
1.403
Min. speed: 80.76 mm/s
Two-action
2.217
Min. speed: 77.85 mm/s
Two-step planner
10.080
Min. speed: 46.04 mm/s
J-NNC loss reduction
86.1% vs two-step
36.7% vs two-action
Same ±10 mm window
0
5
10
Speed deficit (mm)